H4. The Medication Safety Second Pair of Eyes

Real-world pain: A patient in a district hospital is on medicines prescribed by three different doctors across two years. Nobody has the full list. Dangerous interactions, duplicate therapies, and doses unsafe for the patient's kidney function slip through — especially for elderly patients on 8+ drugs.

The problem: Given a patient's consolidated medication list and basic clinical context, flag safety risks with a clear severity and a specific suggested action for the prescriber.

What we provide:

- A simplified drug reference of ~500 medicines: generic name, class, common brands, renal-adjustment flag, pregnancy category
- A curated interaction table (~2,000 pairs) with severity
- 5,000 synthetic patient records: age, weight, eGFR (kidney function), pregnancy status, diagnoses, and a medication list including handwritten-style brand/generic mix and misspellings
- Hidden labels: the set of true safety issues per patient, and 300 deliberately seeded near-miss cases

Assumptions students may make:

- The medication list may be incomplete — the system must not claim the list is safe, only that no issue was found in what it saw
- Brand-to-generic mapping must be resolved before checking (this is much of the real difficulty)
- Reference tables are authoritative and complete for this exercise
- No allergy history is available unless present in the record

Desired output:

1. A prioritised alert list per patient: issue type (interaction / duplicate therapy / renal dose / contraindication), severity, the drugs involved, and a concrete suggested action
2. **Alert fatigue is the real enemy and is scored.** `evaluate.py` measures recall on serious issues *and* penalises low-severity alert volume. A system that flags everything scores near zero — this is the central design lesson
3. A fuzzy brand/generic resolution module with a reported match accuracy on the misspelled inputs
4. Explicit "confidence and limitations" text shown with every alert, plus a mandatory disclaimer that a clinician decides
5. Recall on the 300 seeded near-misses, reported prominently

Stretch: a deprescribing suggestion for polypharmacy patients, with the reasoning chain shown.